Variational Methods Applied to Modelling

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Contents

1. Course introduction
2. Sobolev spaces
3. Elliptic problems
4. Variational approximation
5. $\mathbb{P}_1$ finite element implementation
6. Finite element error estimates

IV. Variational approximation : inner approximation

Let V_h be a **finite-dimensional** vector subspace such that $V_h \to V$ when $h \to 0$.

We replace the variational formulation

$$\text{(FV)} \qquad \text{Find } u \in V \text{ such that } a(u, v) = \ell(v), \ \forall v \in V.$$

by the approximate variational formulation

$$(\text{F}V_h) \qquad \text{Find } u_h \in V_h \text{ such that } a(u_h, v_h) = \ell(v_h), \ \forall v_h \in V_h.$$

With a relevant choice of V_h, we will prove that $u_h \to u$ when $h \to 0$.

We assume that $V_h \subset V$ (**inner approximation**).

IV. Variational approximation : inner approximation

Approximate variational formulation :

$$(\mathrm{F}V_h) \qquad \text{Find } u_h \in V_h \text{ such that } a(u_h, v_h) = \ell(v_h),\ \forall v_h \in V_h.$$

By L. M. theorem, there exists a unique solution $u_h \in V_h$ to $(\mathrm{F}V_h)$:

- $(V_h, \|\cdot\|_V)$ is a Hilbert space (closed vector subspace of V since it is a finite dimensional vector subspace of V).
- Since $V_h \subset V$, the bilinear form a in $V_h \times V_h$ is continuous and coercive and the linear form l in V_h is continuous (with constants M, α and C that do not depend on h).

IV. Variational approximation : inner approximation

Approximate variational formulation :

$$(\mathrm{F}V_h) \qquad \text{Find } u_h \in V_h \text{ such that } a(u_h, v_h) = \ell(v_h),\ \forall v_h \in V_h.$$

Let $(\phi_1, \cdots, \phi_N)$ be a basis of V_h (with $N = \dim V_h$).

We decompose u_h into this basis

$$u_h = \sum_{j=1}^{N} u_h^j \phi_j.$$

Using the linearity of a, it gives

$$\text{Find } (u_h^1, \cdots, u_h^N) \in \mathbb{R}^N \text{ such that } \sum_{j=1}^{N} a(\phi_j, v_h) u_h^j = \ell(v_h),\ \forall v_h \in V_h.$$

IV. Variational approximation : inner approximation

$$\text{Find } (u_h^1, \cdots, u_h^N) \in \mathbb{R}^N \text{ such that } \sum_{j=1}^{N} a(\phi_j, v_h) u_h^j = \ell(v_h), \ \forall v_h \in V_h.$$

is equivalent to

$$\text{Find } (u_h^1, \cdots, u_h^N) \in \mathbb{R}^N \text{ such that } \sum_{j=1}^{N} a(\phi_j, \phi_i) u_h^j = \ell(\phi_i), \ \forall i = 1, \cdots, N.$$

Idea :

$\Rightarrow$: equality true for all $v_h \in V_h$ thus in particular for $\phi_i \in V_h$.

$\Leftarrow$: Since a and l are linear, equality true for all ϕ_i implies equality true for $\sum_{i=1}^{N} v_h^i \phi_i$ that is the decomposition of v_h into the basis $(\phi_1, \cdots, \phi_N)$.

IV. Variational approximation : inner approximation

Find $(u_h^1, \cdots, u_h^N) \in \mathbb{R}^N$ such that $\displaystyle\sum_{j=1}^{N} a(\phi_j, \phi_i) u_h^j = \ell(\phi_i), \ \forall i = 1, \cdots, N.$

The problem writes as a **linear system**

$$\text{Find } U \in \mathbb{R}^N, \qquad AU = L$$

with

$$A_{ij} = a(\phi_j, \phi_i), \quad \forall\, 1 \leq i \leq N, \quad \forall\, 1 \leq j \leq N,$$
$$L_i = l(\phi_i), \quad \forall\, 1 \leq i \leq N.$$

Proposition :
The matrix A is **positive definite** thus invertible.

Remark : Another way to prove it exists a unique solution $u_h \in V_h$ to $(\mathrm{F}V_h)$

IV. Variational approximation : inner approximation

$$\text{(FV)} \qquad \text{Find } u \in V \text{ such that } a(u,v) = \ell(v),\ \forall v \in V.$$

$$\text{(F}V_h\text{)} \qquad \text{Find } u_h \in V_h \text{ such that } a(u_h,v_h) = \ell(v_h),\ \forall v_h \in V_h.$$

Lemma (Céa) :
There exists $C > 0$, independent of h, such that

$$\|u - u_h\|_V \leq C \inf_{v_h \in V_h} \|u - v_h\|_V.$$

The convergence of the inner variational approximation is reduced to a problem of approximation of functions of V by functions of V_h.

IV. Variational approximation : inner approximation

$$\text{(FV)} \qquad \text{Find } u \in V \text{ such that } a(u,v) = \ell(v),\ \forall v \in V.$$

$$\text{(F}V_h\text{)} \qquad \text{Find } u_h \in V_h \text{ such that } a(u_h,v_h) = \ell(v_h),\ \forall v_h \in V_h.$$

Proof :

Since $V_h \subset V$, (FV) gives

$$a(u, w_h) = \ell(w_h),\ \forall w_h \in V_h$$

Consequently,

$$a(u - u_h, w_h) = 0,\ \forall w_h \in V_h$$

Then,

$$a(u-u_h, u-u_h) = a(u-u_h, u-v_h) + a(u-u_h, v_h-u_h) = a(u-u_h, u-v_h).$$

With the coerciveness and the continuity of a, we obtain, for all $v_h \in V_h$,

$$\alpha\|u-u_h\|_V^2 \leq a(u-u_h, u-u_h) = a(u-u_h, u-v_h) \leq M\|u-u_h\|_V\|u-v_h\|_V.$$

IV. Variational approximation : inner approximation

It is time to make clear the meaning of "$V_h \to V$ when $h \to 0$" :

Proposition :

Let $\mathcal{V}$ be a dense subspace of V and $r_h : \mathcal{V} \to V_h$ an operator such that

$$\forall v \in \mathcal{V}, \quad \lim_{h \to 0} \|v - r_h v\|_V = 0,$$

then the solution u_h of (FV_h) tends to the solution u of (FV) when $h \to 0$.

Proof : Let $\epsilon > 0$ and $u \in V$ the solution of (FV).

Since $\mathcal{V}$ is dense in V, $\quad \exists u_\epsilon \in \mathcal{V}, \quad \|u - u_\epsilon\|_V \leq \frac{\epsilon}{2}$.

But, $\lim_{h \to 0} \|u_\epsilon - r_h u_\epsilon\|_V = 0$ i.e. $\exists h_0 > 0, \forall h \in]0, h_0[, \|u_\epsilon - r_h u_\epsilon\|_V \leq \frac{\epsilon}{2}$.

Thus,

$$\exists h_0 > 0, \quad \forall h \in]0, h_0[, \quad \|u - r_h u_\epsilon\|_V \leq \|u - u_\epsilon\|_V + \|u_\epsilon - r_h u_\epsilon\|_V \leq \epsilon.$$

We can use Céa's lemma to conclude

$$\exists h_0 > 0, \ \forall h \in]0, h_0[, \quad \|u - u_h\|_V \leq C \inf_{v_h \in V_h} \|u - v_h\|_V \leq C\|u - r_h u_\epsilon\|_V \leq C\epsilon.$$

IV. Variational approximation : inner approximation

In the following, the objective will be to choose an approximate space $V_h \subset V$ that gives a good compromise between

- an easy-to-solve linear system (sparse matrix with good conditionning),
- a fast convergence speed (with respect to h).

IV. Variational approximation : Hilbert basis

An **Hilbert basis** of an Hilbert space V is an orthonormal countable set $(e_n)_{n\geq 1}$ such that its linear span is dense in V.

If V is a separable Hilbert space, that is there exists a countable set dense in V, then it exists an Hilbert basis.

To build it : from the countable set dense in V, we eventually delete some vectors and renumber them to get a linear independent set and then we use the Gram-Schmidt process.

IV. Variational approximation : Hilbert basis

We choose an Hilbert basis $(e_n)_{n\geq 1}$ and define, for $h = \frac{1}{m}$, $V_h = \text{span}(e_1, e_2, \ldots, e_m)$.

Proposition :
For any integer $m \in \mathbb{N}^*$, there is a unique solution $u_m \in V_h$ to (FV_h) and we have

$$\lim_{m\to+\infty} \|u - u_m\| = 0.$$

Proof : We use the previous result (slide 10) considering for $\mathcal{V}$ the linear span of the set $(e_n)_{n\geq 1}$ (dense in V by definition of the Hilbert basis) and for r_h the orthogonal projector on V_h that verifies

$$\forall v \in \mathcal{V},\ \lim_{h\to 0} \|v - r_h v\|_V = 0.$$

IV. Variational approximation : an example using the monomial basis

1D example : We consider the problem :

$$\begin{cases} -u'' + u = 1, & \text{in }]0,1[, \\ u(0) = 0, u'(1) = -1. \end{cases}$$

Exact solution : $u(x) = 1 - \cosh x + \frac{-1+\sinh 1}{\cosh 1} \sinh x$.

Variational formulation : Defining $V = \left\{u \in H^1(0,1),\ u(0) = 0\right\}$,

$$\text{Find } u \in V, \int_0^1 (u'v' + uv)dx = \int_0^1 v dx - v(1), \quad \forall v \in V$$

Discrete variational formulation : We choose as basis $(x^j)_{1 \leq j \leq m}$.
Find $u_m = \sum_{j=1}^m u_j x^j \in V_{\frac{1}{m}}$,

$$\sum_{j=1}^m \Big(\int_0^1 (ijx^{i+j-2} + x^{i+j})dx \Big) u_j = \int_0^1 x^i dx - 1^i, \quad \forall i = 1, \ldots, m.$$

IV. Variational approximation : an example using the monomial basis

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Discrete variational formulation : We choose as basis $(x^j)_{1 \leq j \leq m}$.
Find $u_m = \sum_{j=1}^m u_j x^j \in V_{\frac{1}{m}}$,

$$\sum_{j=1}^m \left(\frac{ij}{i+j-1} + \frac{1}{i+j+1}\right) u_j = \frac{1}{i+1} - 1, \quad \forall i = 1, \ldots, m.$$

IV. Variational approximation : an example using the monomial basis

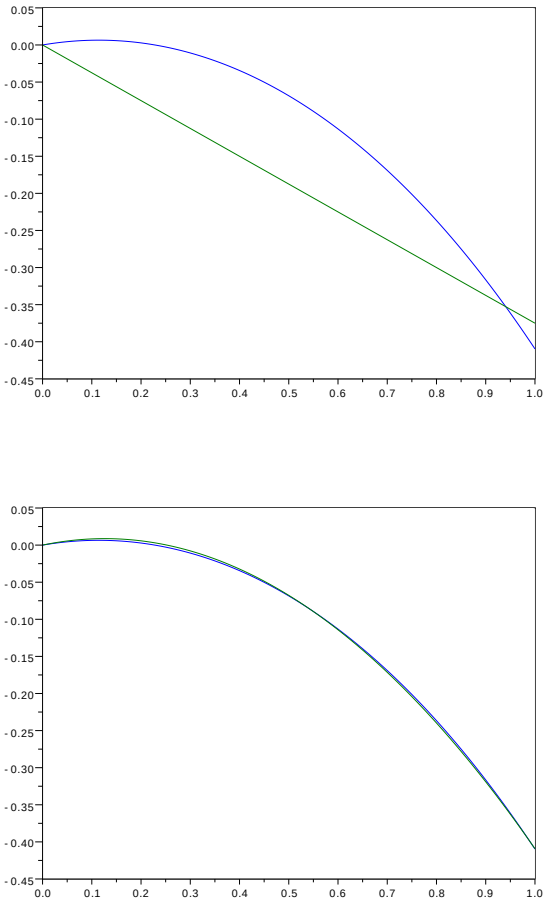

Figure – u (blue) and u_m (green) for $m = 1$ (top) and $m = 2$ (bottom)

m	$\|u - u_m\|_\infty / \|u\|_\infty$
1	0.2909501
2	0.0077572
3	0.0013430
4	0.0000246
5	0.0000027
6	3.539D-08

Table – Convergence table

Unfortunately, for $m = 7$, the condition number of the associated matrix is already of the order of 10^8.

IV. Variational approximation : Hilbert basis

- Such a choice of basis is mostly used **theoretically** to prove existence of PDE solutions (it reduces a problem of infinite dimension to the resolution of a linear system).
- **Not convenient from a numerical point of view** due to the fact we generally obtain an ill-conditioned and full matrix.
- Also, the determination of basis functions verifying the boundary conditions can be difficult for non-academic cases.

IV. Variational approximation : finite element mesh

Idea : For a finite element method (FEM), the finite-dimensional space V_h is defined by **localized basis functions based on a mesh** of the domain.

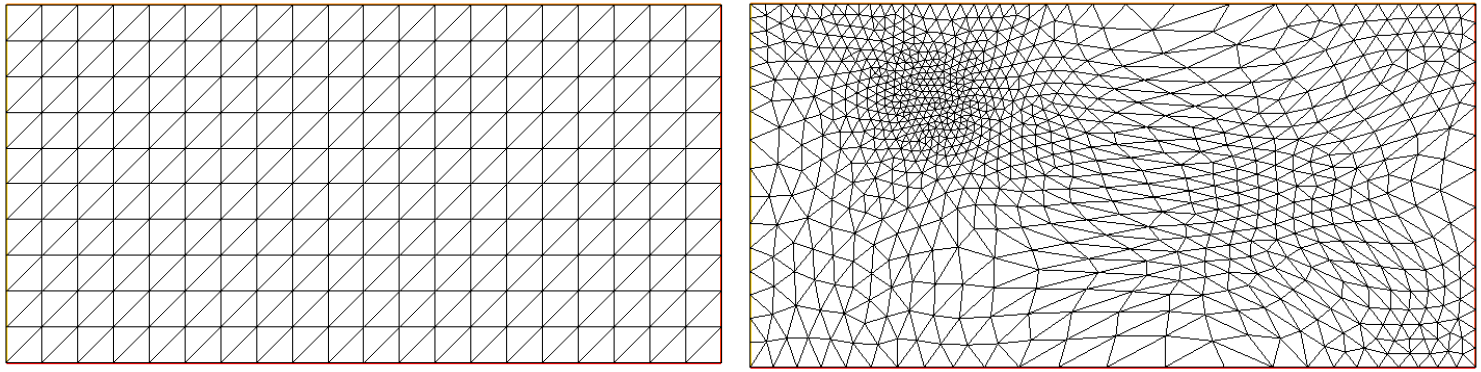

Figure – Examples of triangular meshes of a rectangle (made with Freefem++).

In this context, h is linked to the maximal size of the mesh elements. When h tends to 0, the number of mesh elements N_t and the number of mesh vertices N_v tends to $+\infty$.

Each basis function of V_h will be localized on few adjacent elements generating a sparse matrix.

IV. Variational approximation : finite element mesh

Conform mesh :

- Mesh of the entire domain : $\bigcup_{l=1}^{N_t} \mathcal{T}_l = \overline{\Omega}$
  (with identical elements up to an affine transformation),
- No superposition : $\mathrm{Int}(\mathcal{T}_l) \cap \mathrm{Int}(\mathcal{T}_{l'}) = \emptyset$ if $l \neq l'$,
- No flat element : $\mathrm{Int}(\mathcal{T}_l) \neq \emptyset$,
- An internal edge of an element is always a full internal edge of another.

Triangulation of a 2D domain :

For a triangle, we define its diameter and its roundness (incircle diameter) :

$$h_{\mathcal{T}_l} = \max_{(\mathbf{x}_1,\mathbf{x}_2)\in\mathcal{T}_l^2} \|\mathbf{x}_1 - \mathbf{x}_2\|, \qquad \rho_{\mathcal{T}_l} = \max_{B_r\subset\mathcal{T}_l}(2r).$$

The ratio $\frac{h_{\mathcal{T}_l}}{\rho_{\mathcal{T}_l}} \geq 1$ measure the flatness of $\mathcal{T}_l$ and a good triangulation is a triangulation such that the ratio is not large for any elements.

End of lecture 7.

IV. Variational approximation : $\mathbb{P}_1$ Lagrange finite element

Definition ($\mathbb{P}_1$ finite element) :

$$V_h = \{v_h \in \mathcal{C}^0(\overline{\Omega}),\ v_h|_{\mathcal{T}_l} \in \mathcal{P}_1,\ \forall\ 1 \leq l \leq N_t\}$$

where $\mathcal{P}_1$ is the set of polynomial functions of degree less or equal to 1.

In 2d : $\mathcal{P}_1 = span(1, x, y)$ and $\dim \mathcal{P}_1 = 3$.

Proposition :

$$V_h \subset H^1(\Omega).$$

Indeed, we remind that, for a partition of Ω in M subsets Ω_i with regular boundaries, if $v \in \mathcal{C}^0(\overline{\Omega})$ and $v_l = v_{|\Omega_l} \in H^1(\Omega_i)$ for all l then $v \in H^1(\Omega)$.

IV. Variational approximation : $\mathbb{P}_1$ Lagrange finite element

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where $\mathcal{P}_1$ is the set of polynomial functions of degree less or equal to 1.

In 2d : $\mathcal{P}_1 = span(1, x, y)$ and $\dim \mathcal{P}_1 = 3$.

Proposition : Let $(\mathbf{a}_i)$ be the N_v mesh vertices. Then, for all $v_h, w_h \in V_h$,

$$v_h = w_h \qquad \Leftrightarrow \qquad v_h(\mathbf{a}_i) = w_h(\mathbf{a}_i),\ \forall\ 1 \leq i \leq N_v.$$

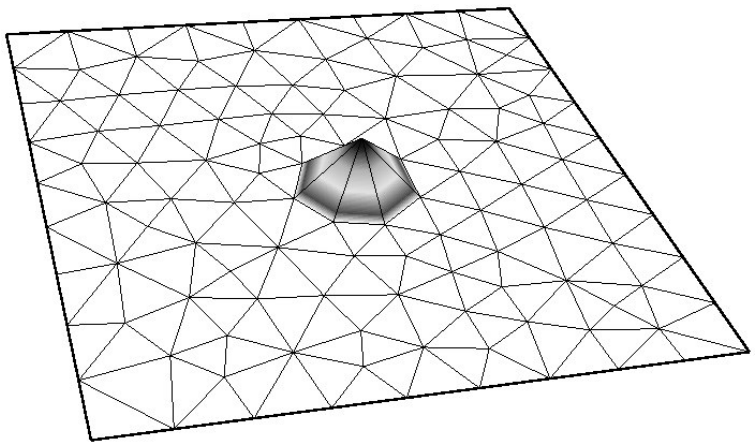
$\Rightarrow$: Equality for all the points of Ω thus in particular for all vertices $\mathbf{a}_i$.
$\Leftarrow$: $v_h(\mathbf{a}_i) = w_h(\mathbf{a}_i)\ \forall i \Rightarrow v_h|_{\mathcal{T}_l} = w_h|_{\mathcal{T}_l}\ \forall l$ and $\bigcup_l \mathcal{T}_l = \overline{\Omega}$.

IV. Variational approximation : $\mathbb{P}_1$ Lagrange finite element :

Definition ($\mathbb{P}_1$ finite element) :

$$V_h = \{v_h \in \mathcal{C}^0(\overline{\Omega}),\ v_h|_{\mathcal{T}_l} \in \mathcal{P}_1,\ \forall\ 1 \leq \cdots \leq N_t\}.$$

Proposition : V_h is a vector space of dimension N_v and a basis of V_h is given by functions $\phi_i \in V_h$ defined by $\phi_i(\mathbf{a}_j) = \delta_{ij}$ for all $1 \leq i, j \leq N_v$.



$$\text{supp}(\phi_i) = \bigcup_{l, \mathbf{a}_i \in \mathcal{T}_l} \mathcal{T}_l$$

Figure – 2d finite element $\mathbb{P}_1$ (global) basis function also called "hat function" [picture coming from Allaire's book, 06].

IV. Variational approximation : $\mathbb{P}_1$ Lagrange finite element :

Definition ($\mathbb{P}_1$ finite element) :

$$V_h = \{v_h \in \mathcal{C}^0(\overline{\Omega}),\ v_h|_{\mathcal{T}_l} \in \mathcal{P}_1,\ \forall\ 1 \leq \cdots \leq N_t\}.$$

Proposition : V_h is a vector space of dimension N_v and a basis of V_h is given by functions $\phi_i \in V_h$ defined by $\phi_i(\mathbf{a}_j) = \delta_{ij}$ for all $1 \leq i, j \leq N_v$.

- Linearly independent set :

$$\sum_{i=1}^{N_v} \alpha_i \phi_i(\mathbf{x}) = 0\ \forall \mathbf{x} \in \overline{\Omega} \quad \Rightarrow \quad \sum_{i=1}^{N_v} \alpha_i \phi_i(\mathbf{a}_j) = \alpha_j = 0\ \forall 1 \leq j \leq N_v.$$

- Spanning V_h : We have to show that $V_h \subset span(\phi_1, \cdots, \phi_{N_v})$. For all $v_h \in V_h$, $v_h(\mathbf{a}_j) = \sum_{i=1}^{N_v} v_h(\mathbf{a}_i)\phi_i(\mathbf{a}_j)\ \forall j$. Thus,

$$v_h(\mathbf{x}) = \sum_{i=1}^{N_v} v_h(\mathbf{a}_i)\phi_i(\mathbf{x})$$

and consequently $v_h \in span(\phi_1, \cdots, \phi_{N_v})$.

IV. Variational approximation : $\mathbb{P}_1$ Lagrange finite element :

Homogeneous Dirichlet boundary conditions :

Problem :

$$V_h = \{v_h \in \mathcal{C}^0(\overline{\Omega}),\ v_h|_{\mathcal{T}_l} \in \mathcal{P}_1,\ \forall\ 1 \leq \cdots \leq N_t\} \not\subset H_0^1(\Omega).$$

Solution :

$$\begin{aligned} \widetilde{V}_h = V_h \cap H_0^1(\Omega) &= \{v_h \in V_h,\ v_h(\mathbf{x}) = 0\ \forall\ \mathbf{x} \in \partial\Omega\} \\ &= \{v_h \in V_h,\ v_h(\mathbf{a}_i) = 0\ \forall\ \mathbf{a}_i \in \partial\Omega\} \\ &= \text{span}\{\phi_i, \mathbf{a}_i \notin \partial\Omega\}. \end{aligned}$$

We consider only the basis functions associated to the $N_{v,int}$ internal vertices and thus

$$dim\widetilde{V}_h = N_{v,int}.$$

IV. Variational approximation : finite elements

General definition :

A **finite element** is a triple $(K, \mathcal{P}, \Sigma)$, where

- K is a non empty compact subset $\mathbb{R}^d$
- $\mathcal{P}$ is a finite dimensional vector space of functions from K to $\mathbb{R}$.
- $\Sigma = \{\ell_1, \ell_2, \ldots, \ell_m\}$ is a set of m linear forms on $\mathcal{P}$. The reals $\ell_j(p)$ are called the **local degrees of freedom**.

Example : Reference $\mathbb{P}_1$ finite element :

- In 1D $(d = 1)$: $K = [0, 1]$, $\mathcal{P} = \mathbb{R}_1[X]$, $\Sigma = \{p \to p(0), p \to p(1)\}$.
- In 2D $(d = 2)$: $K = \{(x, y), x \geq 0, y \geq 0, x + y \leq 1\}$, $\mathcal{P} = \text{span}(1, x, y)$, $\Sigma = \{p \to p(0, 0), p \to p(1, 0), p \to p(0, 1)\}$.

IV. Variational approximation : finite elements

Definition :

The finite element $(K, \mathcal{P}, \Sigma)$ is **unisolvent** when

$$\forall(\alpha_1, \dots, \alpha_m) \in \mathbb{R}^m,\ \exists! p \in \mathcal{P},\ \forall j = 1 \dots m,\ \ell_j(p) = \alpha_j$$

Then, we call **local basis functions** of the finite element the m functions $\phi_i \in \mathcal{P}$, $i = 1, \dots, m$, such that

$$\ell_j(\phi_i) = \delta_{ij}, \quad \forall j = 1, \dots, m.$$

Remark :

- If $\operatorname{card} \Sigma \neq \dim \mathcal{P}$ then the finite element is not unisolvent.
- if $\operatorname{card} \Sigma = \dim \mathcal{P}$, we can study if $0_{\mathcal{P}}$ is the only element of $\mathcal{P}$ such that $\ell_j(p) = 0\ \forall j = 1, \dots, m$.

IV. Variational approximation : finite elements

Definition :
The finite element $(K, \mathcal{P}, \Sigma)$ is **unisolvent** when

$$\forall(\alpha_1, \ldots, \alpha_m) \in \mathbb{R}^m,\ \exists! p \in \mathcal{P},\ \forall j = 1 \ldots m,\ \ell_j(p) = \alpha_j$$

Then, we call **local basis functions** of the finite element the m functions $\phi_i \in \mathcal{P}$, $i = 1, \ldots, m$, such that

$$\ell_j(\phi_i) = \delta_{ij}, \quad \forall j = 1, \ldots, m.$$

Example : The $\mathbb{P}_1$ finite element is unisolvent. The basis functions for the reference element are :

- In 1D $(d = 1)$: $\phi_1(x) = 1 - x$ and $\phi_2(x) = x$.
- In 2D $(d = 2)$: $\phi_1(x, y) = 1 - x - y$, $\phi_2(x, y) = x$ and $\phi_3(x, y) = y$.

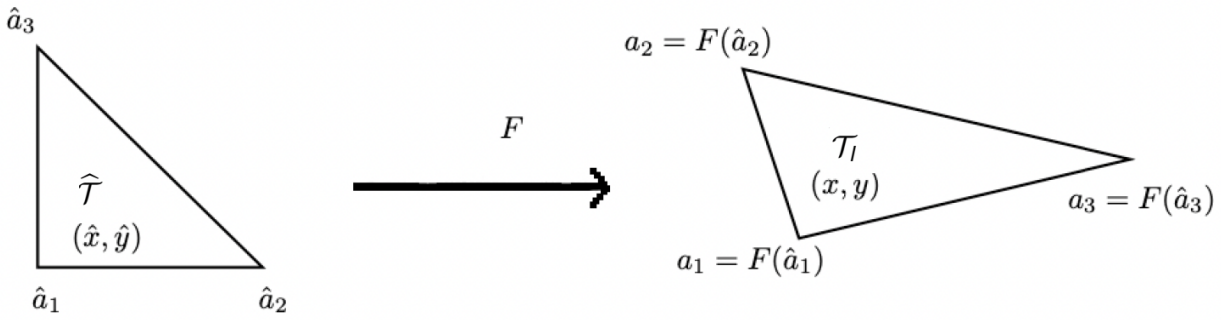
IV. Variational approximation : finite elements

Definition :

Two finite elements $(\hat{K}, \hat{\mathcal{P}}, \hat{\Sigma})$ and $(K, \mathcal{P}, \Sigma)$ are **affine equivalent** when it exists an invertible **affine transformation** F from $\hat{K}$ to K ($F : \hat{\mathbf{x}} \to B\hat{\mathbf{x}} + b$ with $B \in GL_d(\mathbb{R})$ and $b \in \mathbb{R}^d$) such that :

- $K = F(\hat{K})$,
- $\mathcal{P} = \left\{ p : K \to \mathbb{R},\ p \circ F \in \hat{\mathcal{P}} \right\}$,
- $\Sigma = \left\{ \ell_j : \mathcal{P} \to \mathbb{R} \text{ for } j = 1, \dots, m, \quad \ell_j(p) = \hat{\ell}_j(p \circ F),\ \hat{\ell}_j \in \hat{\Sigma} \right\}$.

Example : affine transformation for triangles in 2D :



$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \end{pmatrix} \begin{pmatrix} \widehat{x} \\ \widehat{y} \end{pmatrix} + \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} := B \begin{pmatrix} \widehat{x} \\ \widehat{y} \end{pmatrix} + b = F \begin{pmatrix} \widehat{x} \\ \widehat{y} \end{pmatrix}.$$

IV. Variational approximation : finite elements

Proposition :

Local basis functions ϕ_i of $(K, \mathcal{P}, \Sigma)$ are expressed by the local basis functions $\hat{\phi}_i$ of $(\hat{K}, \hat{\mathcal{P}}, \hat{\Sigma})$ via the affine transformation F :

$$\phi_i = \hat{\phi}_i \circ F^{-1}, \quad \forall i = 1, \dots, m.$$

Example : $\mathbb{P}_1$ finite element in 1D :

Affine transformation :

$$x = F(\hat{x}) = x_i + \hat{x}(x_{i+1} - x_i), \quad \hat{x} = F^{-1}(x) = \frac{x - x_i}{x_{i+1} - x_i}.$$

Local reference basis functions defined on $\hat{K} = [0, 1]$:

$$\hat{\phi}_1(\hat{x}) = 1 - \hat{x}, \quad \hat{\phi}_2(\hat{x}) = \hat{x}$$

Local basis functions defined on $K = [x_i, x_{i+1}]$:

$$\phi_1(x) = \frac{x - x_{i+1}}{x_i - x_{i+1}}, \quad \phi_1(x) = \frac{x - x_i}{x_{i+1} - x_i}$$

IV. Variational approximation : finite elements

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In practise, we work with a **set of affine equivalent finite elements** $(K_l, \mathcal{P}_l, \Sigma_l)$. All integral computations are reduced to a computation on the **reference element** $(\hat{K}, \hat{\mathcal{P}}, \hat{\Sigma})$.

IV. Variational approximation : finite elements

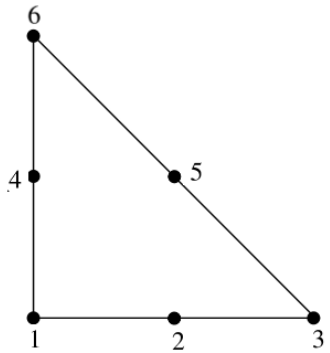
Different finite element examples :

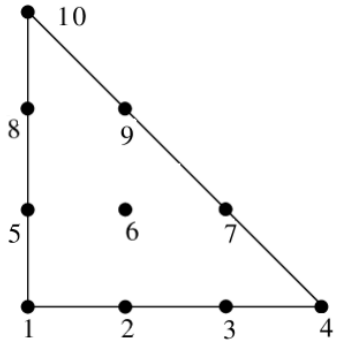
For Lagrange finite elements (or nodal elements), each degree of freedom corresponds to the value of the function on a corresponding node.

- $\mathbb{P}_2$ in 1D :

$K = [0,1]$, $\mathcal{P} = \text{span}(1, x, x^2)$, $\Sigma = \{p \to p(0), p \to p(1/2), p \to p(1)\}$
$\phi_1(x) = 2(x - \frac{1}{2})(x-1)$, $\phi_2(x) = 4x(x-1)$ and $\phi_3(x) = 2x(x - \frac{1}{2})$.

- $\mathbb{P}_2$ and $\mathbb{P}_3$ in 2D :





Global level : $V_h^k = \{v_h \in \mathcal{C}^0(\overline{\Omega}),\ v_h|_{\mathcal{T}_l} \in \mathcal{P}_k,\ \forall\ 1 \leq \cdots \leq N_t\}$.

$$dim V_h^1 = N_v, \quad dim V_h^2 = N_v + N_{ed}, \quad dim V_h^3 = N_v + 2N_{ed} + N_t.$$

IV. Variational approximation : finite elements

Different finite element examples :

$\mathbb{P}_k$:

$$V_h^k = \{v_h \in \mathcal{C}^0(\overline{\Omega}),\ v_h|_{\mathcal{T}_l} \in \mathcal{P}_k,\ \forall\ 1 \leq \cdots \leq N_t\}.$$

Increasing k,

- the matrix size increases,
- the matrix stays sparse but the number of non-zero terms increases,
- the approximation is a priori better (assuming the solution is regular enough)

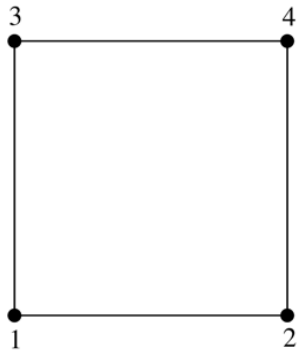
$$\|u - u_h\|_{H^1(\Omega)} \leq Ch^k \|u\|_{H^{k+1}(\Omega)}.$$

IV. Variational approximation : finite elements

Different finite element examples :

- $\mathbb{Q}_1$ in 2D :

$K = [0,1]^2$, $\mathcal{P} = \text{span}(1, x, y, xy)$,
$\Sigma = \{p \to p(0,0), p \to p(1,0), p \to p(0,1), p \to p(1,1)\}$



Local basis functions : $\phi_1(x,y) = (1-x)(1-y)$, $\phi_2(x,y) = x(1-y)$, $\phi_3(x,y) = (1-x)y$ and $\phi_4(x,y) = xy$.

Tensor products of 1D finite elements but limited to simple geometries.

IV. Variational approximation : finite elements

Different finite element examples :

For Hermite finite elements, each degree of freedom corresponds to the value of the function and its derivatives on a corresponding node.

• cubic Hermite in 1D :
$K = [0,1]$, $\mathcal{P} = \mathbb{R}_3[X]$, $\Sigma = \{p \to p(0), p \to p'(0), p \to p(1), p \to p'(1)\}$

$\phi_1(x) = 1 - 3x^2 + 2x^3$, $\phi_2(x) = x - 2x^2 + x^3$, $\phi_3(x) = 3x^2 - 2x^3$ and $\phi_4(x) = -x^2 + x^3$.

Global level : $V_h = \{v_h \in \mathcal{C}^1(\overline{\Omega}),\ v_h|_{[x_i,x_{i+1}]} \in \mathbb{R}_3[X],\ \forall i\} \subset H^2(0,1)$.

$$u_h(x) = \sum_{i=1}^{N_v} u_h(a_i)\phi_{2i-1}(x) + \sum_{i=1}^{N_v} u_h'(a_i)\phi_{2i}(x)$$

Used for instance to solve 4-th order elliptic PDEs.

End of lecture 8.

Contents

1. Course introduction
2. Sobolev spaces
3. Elliptic problems
4. Variational approximation
5. $\mathbb{P}_1$ finite element implementation
6. Finite element error estimates

V. $\mathbb{P}_1$ finite element implementation

We consider the problem

$$\text{Find } u \text{ such that} \qquad \begin{cases} -\Delta u + u = f & \text{in } \Omega, \\ \partial_n u = 0 & \text{on } \partial\Omega. \end{cases}$$

Associated variational formulation :

$$\text{Find } u \in H^1(\Omega), \int_\Omega \nabla u \cdot \nabla v \; dx + \int_\Omega uvdx = \int_\Omega fvdx \quad \forall v \in H^1(\Omega).$$

We use P_1 finite element defining

$$V_h = \{v_h \in \mathcal{C}^0(\overline{\Omega}), \; v_h|_{\mathcal{T}_l} \in \mathcal{P}_1, \; \forall \; 1 \leq \cdots \leq N_t\}.$$

Associated approximate variational formulation :

$$\text{Find } u_h \in V_h, \int_\Omega \nabla u_h \cdot \nabla v_h \; dx + \int_\Omega u_h v_h dx = \int_\Omega f v_h dx \quad \forall v_h \in V_h.$$

V. $\mathbb{P}_1$ finite element implementation

We decompose u_h into the P_1 finite element basis

$$u_h = \sum_{j=1}^{N_v} u_h^j \phi_j.$$

The problem becomes

$$\text{Find } U \in \mathbb{R}^{N_v}, \qquad AU = L$$

with, for $1 \leq i, j \leq N_v$, $U_j = u_h^j$,

$$L_i = l(\phi_i) = \int_\Omega f \phi_i dx$$

and

$$A_{ij} = a(\phi_j, \phi_i) = \int_\Omega \nabla \phi_j \cdot \nabla \phi_i \; dx + \int_\Omega \phi_j \phi_i dx.$$

V. $\mathbb{P}_1$ finite element implementation

We decompose u_h into the P_1 finite element basis

$$u_h = \sum_{j=1}^{N_v} u_h^j \phi_j.$$

The problem becomes

$$\text{Find } U \in \mathbb{R}^{N_v}, \qquad (K+M)U = L$$

where K is the **stiffness matrix**

$$K_{ij} = a(\phi_j, \phi_i) = \int_\Omega \nabla\phi_j \cdot \nabla\phi_i \; dx$$

and M the **mass matrix**

$$M_{ij} = \int_\Omega \phi_j \phi_i dx.$$

V. $\mathbb{P}_1$ finite element implementation : 1D example

For simplicity of the presentation, we consider a uniform discretization : $x_i = (i-1)h$, for all $i = 1, \ldots, N_v$ ($h = \frac{L}{N_v - 1}$).

Basis functions :

$$\phi_i(x) = \begin{cases} \frac{x - x_{i-1}}{h} & \text{if } x \in [x_{i-1}, x_i] \\ -\frac{x - x_{i+1}}{h} & \text{if } x \in [x_i, x_{i+1}] \\ 0 & \text{otherwise} \end{cases}.$$

V. $\mathbb{P}_1$ finite element implementation : 1D example

$$K_{ij} = \int_\Omega \phi_j'(x)\phi_i'(x)\ dx$$

i-th line of the stiffness matrix K :

$$\begin{pmatrix} \int_\Omega \phi_1'\phi_i' & \dots & \int_\Omega \phi_i'\phi_i' & \dots & \int_\Omega \phi_{N_v}'\phi_i' \end{pmatrix}$$

V. $\mathbb{P}_1$ finite element implementation : 1D example

$$K_{ij} = \int_\Omega \phi_j'(x)\phi_i'(x)\ dx$$

Since $\text{supp}(\phi_i \cap \phi_j) = \emptyset$ as soon as $|i-j| \geq 2$, the i-th line of the stiffness matrix has only 3 non-zeros entries :

$$\begin{pmatrix} 0 & \dots & 0 & \int_\Omega \phi_{i-1}'\phi_i' & \int_\Omega \phi_i'\phi_i' & \int_\Omega \phi_{i+1}'\phi_i' & 0 & \dots & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & \dots & 0 & \int_{x_{i-1}}^{x_i} \phi_{i-1}'\phi_i' & \int_{x_{i-1}}^{x_{i+1}} \phi_i'\phi_i' & \int_{x_i}^{x_{i+1}} \phi_i'\phi_{i+1}' & 0 & \dots & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & \dots & 0 & \int_{x_{i-1}}^{x_i} -\frac{1}{h^2} & \int_{x_{i-1}}^{x_{i+1}} \frac{1}{h^2} & \int_{x_i}^{x_{i+1}} -\frac{1}{h^2} & 0 & \dots & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & \dots & 0 & -\frac{1}{h} & \frac{2}{h} & -\frac{1}{h} & 0 & \dots & 0 \end{pmatrix}$$

V. $\mathbb{P}_1$ finite element implementation : 1D example

$$M_{ij} = \int_\Omega \phi_j(x)\phi_i(x)\ dx$$

In the same way, i-th line of the mass matrix M :

$$\begin{pmatrix} 0 & \dots & 0 & \int_{x_{i-1}}^{x_i} \phi_{i-1}\phi_i & \int_{x_{i-1}}^{x_{i+1}} \phi_i\phi_i & \int_{x_i}^{x_{i+1}} \phi_i\phi_{i+1} & 0 & \dots & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & \dots & 0 & \frac{h}{6} & \frac{2h}{3} & \frac{h}{6} & 0 & \dots & 0 \end{pmatrix}$$

V. $\mathbb{P}_1$ finite element implementation : 1D example

i-th component of the second member :

$$L_i = \int_\Omega f\phi_i \, dx = \int_{x_{i-1}}^{x_{i+1}} f\phi_i \, dx.$$

If f does not allow to compute explicitly this integral, a possibility is to approximate f by a function of V_h :

$$f(x) \approx f(x_i)\phi_i + f(x_{i+1})\phi_{i+1} \qquad \text{for } x \in [x_i, x_{i+1}].$$

Then,

$$\begin{aligned} L_i &\approx f(x_{i-1}) \int_{x_{i-1}}^{x_i} \phi_{i-1}\phi_i + f(x_i) \int_{x_{i-1}}^{x_{i+1}} \phi_i^2 + f(x_{i+1}) \int_{x_i}^{x_{i+1}} \phi_{i+1}\phi_i \\ &\approx \frac{h}{6}\Big(f(x_{i-1}) + 4f(x_i) + f(x_{i+1})\Big) \end{aligned}$$

(Simpson's quadrature rule).

V. $\mathbb{P}_1$ finite element implementation : matrix assembling

In 2D :

Loop on vertices
(matrices built line by line) :

$K = 0$
$M = 0$
For $i = 1, \dots, N_v$
    For $j = 1, \dots, N_v$
        $K_{i,j} = \int_\Omega \nabla\phi_j \cdot \nabla\phi_i$
        $M_{i,j} = \int_\Omega \phi_j \phi_i$
    end
end

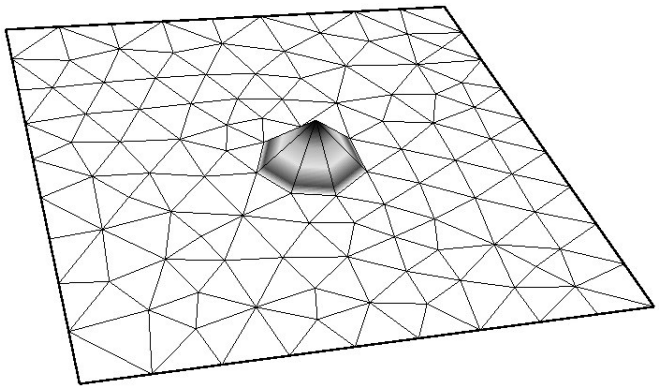


Figure – 2d $\mathbb{P}_1$ (global) basis function ϕ_i.

Pb : A lot of zero terms are computed.

V. $\mathbb{P}_1$ finite element implementation : matrix assembling

In 2D :

Loop on vertices
(matrices built line by line) :

$K = 0$
$M = 0$
For $i = 1, \dots, N_v$
    For $j = 1, \dots, N_v$
        $K_{i,j} = \int_\Omega \nabla\phi_j \cdot \nabla\phi_i$
        $M_{i,j} = \int_\Omega \phi_j \phi_i$
    end
end

Pb : A lot of zero terms are computed.

Loop on vertices
(matrices built line by line) :

$K = 0$
$M = 0$
For $i = 1, \dots, N_v$
    For j such that $\mathbf{a_j}$ neighbour of $\mathbf{a_i}$
        $K_{i,j} = \int_\Omega \nabla\phi_j \cdot \nabla\phi_i$
        $M_{i,j} = \int_\Omega \phi_j \phi_i$
    end
end

Pb : How to determinate the neighbouring vertices ?

V. $\mathbb{P}_1$ finite element implementation : matrix assembling

Mesh information :

Vertex coordinate table

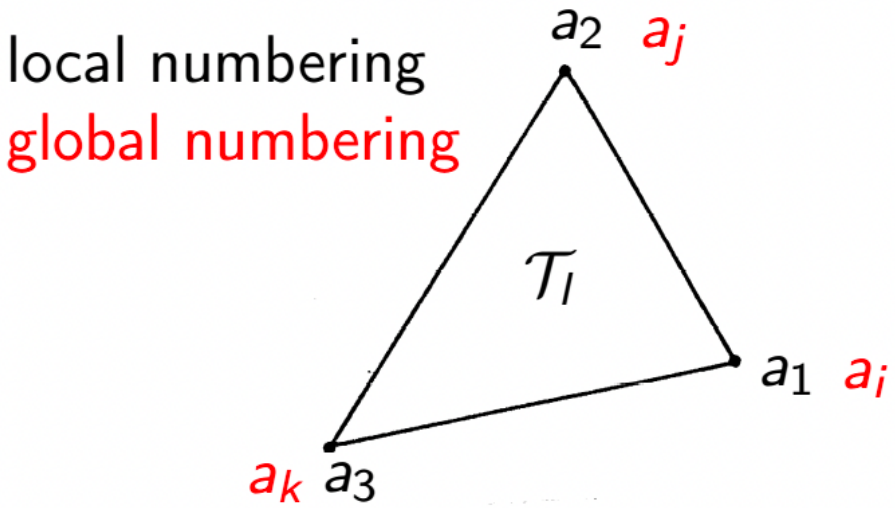
$i = 1, \cdots, N_v$

x_i	y_i

Link Triangle - Vertex table

$l = 1, \cdots, N_t$

i	j	k



V. $\mathbb{P}_1$ finite element implementation : matrix assembling

In 2D :

Loop on triangles $\left(\int_\Omega = \sum_{l=1}^{N_t} \int_{\mathcal{T}_l} \right)$:

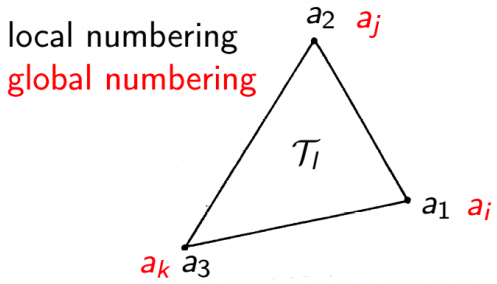
```
K = 0
M = 0
For l = 1, ..., N_t
    Build K^l_elem and M^l_elem

    For ldof1 = 1, ..., 3
        m = T[l, ldof1]
        For ldof2 = 1, ..., 3
            n = T[l, ldof2]
            K[m, n] += K^l_elem[ldof1, ldof2]
            M[m, n] += M^l_elem[ldof1, ldof2]
        end
    end

end
```

We define **elementary matrices** such that

$$M^l_{elem} = \begin{pmatrix} \int_{\mathcal{T}_l} \phi_1\phi_1 & \int_{\mathcal{T}_l} \phi_2\phi_1 & \int_{\mathcal{T}_l} \phi_3\phi_1 \\ \int_{\mathcal{T}_l} \phi_1\phi_2 & \int_{\mathcal{T}_l} \phi_2\phi_2 & \int_{\mathcal{T}_l} \phi_3\phi_2 \\ \int_{\mathcal{T}_l} \phi_1\phi_3 & \int_{\mathcal{T}_l} \phi_2\phi_3 & \int_{\mathcal{T}_l} \phi_3\phi_3 \end{pmatrix}$$

and then we **assemble the global matrices** using the triangle - vertex table T

$$l = 1, \cdots, N_t \quad \begin{array}{|c|c|c|} \hline i & j & k \\ \hline \end{array}$$



V. $\mathbb{P}_1$ finite element implementation : elementary matrix computation

Elementary mass matrix :

$$M^I_{elem} = \begin{pmatrix} \int_{\mathcal{T}_I} \phi_1\phi_1 & \int_{\mathcal{T}_I} \phi_2\phi_1 & \int_{\mathcal{T}_I} \phi_3\phi_1 \\ \int_{\mathcal{T}_I} \phi_1\phi_2 & \int_{\mathcal{T}_I} \phi_2\phi_2 & \int_{\mathcal{T}_I} \phi_3\phi_2 \\ \int_{\mathcal{T}_I} \phi_1\phi_3 & \int_{\mathcal{T}_I} \phi_2\phi_3 & \int_{\mathcal{T}_I} \phi_3\phi_3 \end{pmatrix}$$

Elementary stiffness matrix :

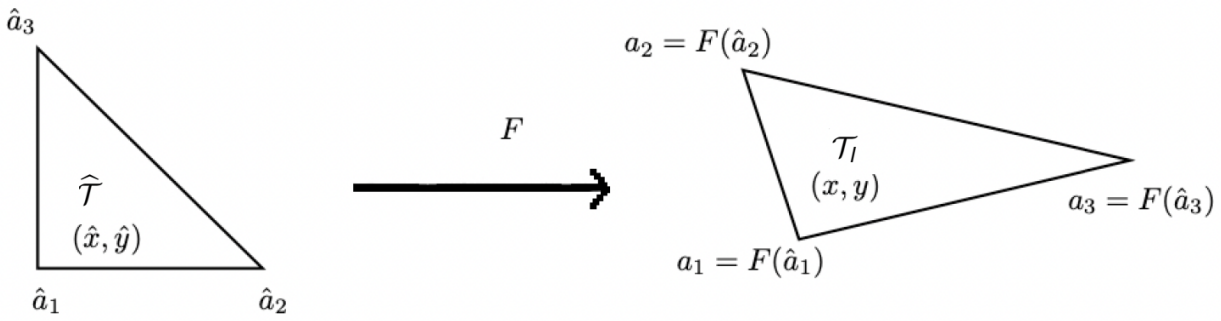
$$K^I_{elem} = \begin{pmatrix} \int_{\mathcal{T}_I} \nabla\phi_1\cdot\nabla\phi_1 & \int_{\mathcal{T}_I} \nabla\phi_2\cdot\nabla\phi_1 & \int_{\mathcal{T}_I} \nabla\phi_3\cdot\nabla\phi_1 \\ \int_{\mathcal{T}_I} \nabla\phi_1\cdot\nabla\phi_2 & \int_{\mathcal{T}_I} \nabla\phi_2\cdot\nabla\phi_2 & \int_{\mathcal{T}_I} \nabla\phi_3\cdot\nabla\phi_2 \\ \int_{\mathcal{T}_I} \nabla\phi_1\cdot\nabla\phi_3 & \int_{\mathcal{T}_I} \nabla\phi_2\cdot\nabla\phi_3 & \int_{\mathcal{T}_I} \nabla\phi_3\cdot\nabla\phi_3 \end{pmatrix}$$

Elementary second member :

$$L^I_{elem} = \begin{pmatrix} \int_{\mathcal{T}_I} f\phi_1 \\ \int_{\mathcal{T}_I} f\phi_2 \\ \int_{\mathcal{T}_I} f\phi_3 \end{pmatrix}$$

V. $\mathbb{P}_1$ finite element implementation : elementary matrix computation

To compute each entries, we work on the **reference element** by making a change of variable :



$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \end{pmatrix} \begin{pmatrix} \widehat{x} \\ \widehat{y} \end{pmatrix} + \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} := B \begin{pmatrix} \widehat{x} \\ \widehat{y} \end{pmatrix} + b = F \begin{pmatrix} \widehat{x} \\ \widehat{y} \end{pmatrix}.$$

Then,

$$\begin{aligned} \int_{\mathcal{T}_l} w(x,y)dxdy &= \int_{\widehat{\mathcal{T}}} w\big(F(\widehat{x},\widehat{y})\big)|\det(B)|d\widehat{x}d\widehat{y} \\ &= 2\text{Area}(\mathcal{T}_l) \int_{\widehat{\mathcal{T}}} w\big(F(\widehat{x},\widehat{y})\big)\ d\widehat{x}d\widehat{y}. \end{aligned}$$

V. $\mathbb{P}_1$ finite element implementation : elementary matrix computation

For the reference triangle $\widehat{\mathcal{T}}$, the basis functions have simple expressions :

$$\widehat{\phi}_1(\widehat{x},\widehat{y}) = 1-\widehat{x}-\widehat{y}, \qquad \widehat{\phi}_2(\widehat{x},\widehat{y}) = \widehat{x} \quad \text{and} \quad \widehat{\phi}_3(\widehat{x},\widehat{y}) = \widehat{y}.$$

Computation examples :

$$\begin{aligned}
\int_{\mathcal{T}^l} \phi_2\phi_3 dxdy &= 2\text{Area}(\mathcal{T}^l)\int_{\widehat{\mathcal{T}}} \widehat{\phi}_2\widehat{\phi}_3 d\widehat{x}d\widehat{y} \\
&= 2\text{Area}(\mathcal{T}^l)\int_0^1 \widehat{x}\Big(\int_0^{1-\widehat{x}} \widehat{y}d\widehat{y}\Big)d\widehat{x} \\
&= \text{Area}(\mathcal{T}^l)\int_0^1 \widehat{x}(1-\widehat{x})^2 d\widehat{x} \\
&= \text{Area}(\mathcal{T}^l)\int_0^1 (\widehat{x}-2\widehat{x}^2+\widehat{x}^3)d\widehat{x} \\
&= \text{Area}(\mathcal{T}^l)\Big(\frac{1}{2}-\frac{2}{3}+\frac{1}{4}\Big) = \frac{\text{Area}(\mathcal{T}^l)}{12}.
\end{aligned}$$

V. $\mathbb{P}_1$ finite element implementation : elementary matrix computation

Computation examples :

$$
\begin{aligned}
\int_{\mathcal{T}^l} \left|\nabla \phi_2\right|^2 dxdy &= \int_{\mathcal{T}^l} \left[\left(\frac{\partial \phi_2}{\partial x}\right)^2 + \left(\frac{\partial \phi_2}{\partial y}\right)^2\right] dxdy \\
&= 2\mathrm{Area}(\mathcal{T}^l) \int_{\widehat{\mathcal{T}}} \left[\left(\frac{\partial \widehat{\phi}_2}{\partial \widehat{x}}\frac{\partial \widehat{x}}{\partial x} + \frac{\partial \widehat{\phi}_2}{\partial \widehat{y}}\frac{\partial \widehat{y}}{\partial x}\right)^2 + \left(\frac{\partial \widehat{\phi}_2}{\partial \widehat{x}}\frac{\partial \widehat{x}}{\partial y} + \frac{\partial \widehat{\phi}_2}{\partial \widehat{y}}\frac{\partial \widehat{y}}{\partial y}\right)^2\right] d\widehat{x}d\widehat{y} \\
&= 2\mathrm{Area}(\mathcal{T}^l) \int_{\widehat{\mathcal{T}}} \left[\left(\frac{\partial \widehat{x}}{\partial x}\right)^2 + \left(\frac{\partial \widehat{x}}{\partial y}\right)^2\right] d\widehat{x}d\widehat{y} \\
&= \frac{1}{2\mathrm{Area}(\mathcal{T}^l)} \int_{\widehat{\mathcal{T}}} \left[(y_3 - y_1)^2 + (x_1 - x_3)^2\right] d\widehat{x}d\widehat{y} \\
&= \frac{\mathrm{Area}(\widehat{\mathcal{T}})}{2\mathrm{Area}(\mathcal{T}^l)} \left((y_3 - y_1)^2 + (x_1 - x_3)^2\right) \\
&= \frac{(y_3 - y_1)^2 + (x_1 - x_3)^2}{4\mathrm{Area}(\mathcal{T}^l)}.
\end{aligned}
$$

V. $\mathbb{P}_1$ finite element implementation : elementary matrix computation

For other computations, we obtain :

$$\int_{\mathcal{T}^l} \nabla\phi_n \cdot \nabla\phi_{n'} dxdy \quad = \quad \frac{b_{n1}b_{n'1} + b_{n2}b_{n'2}}{4\text{Area}(\mathcal{T}^l)}, \quad \text{for } n, n' = 1, 2 \text{ or } 3$$

with

$$b_{11} = y_2 - y_3, \qquad b_{12} = x_3 - x_2,$$

$$b_{21} = y_3 - y_1, \qquad b_{22} = x_1 - x_3,$$

$$b_{31} = y_1 - y_2, \qquad b_{32} = x_2 - x_1.$$

End of lecture 9.

V. $\mathbb{P}_1$ finite element implementation : barycentric coordinates

$$\forall m \in \mathcal{T}^I, \quad \begin{cases} \lambda_1(m)\overrightarrow{Oa_1} + \lambda_2(m)\overrightarrow{Oa_2} + \lambda_3(m)\overrightarrow{Oa_3} = \overrightarrow{Om} \\ \lambda_1(m) + \lambda_2(m) + \lambda_3(m) = 1 \end{cases}.$$

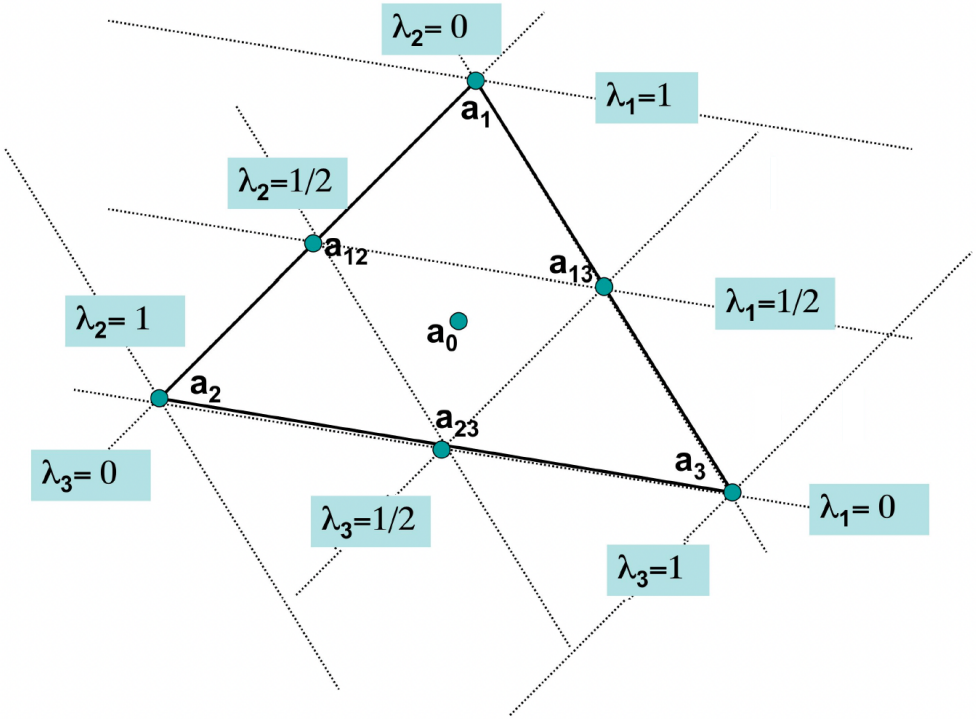


Figure – Barycentric coordinate of a triangle (picture from an E. Blayo's course).

V. $\mathbb{P}_1$ finite element implementation : barycentric coordinates

$$\forall (x,y) \in \mathcal{T}^l, \quad \begin{cases} x_1\lambda_1(x,y) + x_2\lambda_2(x,y) + x_3\lambda_3(x,y) = x \\ y_1\lambda_1(x,y) + y_2\lambda_2(x,y) + y_3\lambda_3(x,y) = y \\ \lambda_1(x,y) + \lambda_2(x,y) + \lambda_3(x,y) = 1 \end{cases} .$$

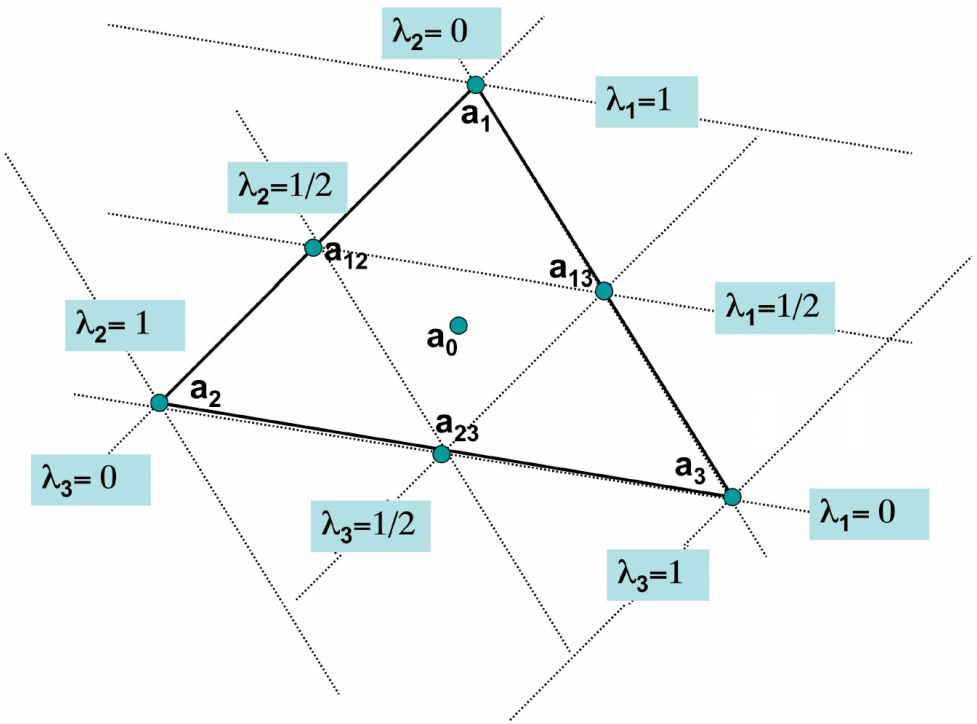


Figure – Barycentric coordinate of a triangle (picture from an E. Blayo's course).

V. $\mathbb{P}_1$ finite element implementation : barycentric coordinates

Proposition :
Local $\mathbb{P}_1$ basis functions are the barycentric coordinates :

$$\phi_i(x,y) = \lambda_i(x,y), \quad i = 1,\dots,3.$$

Proof : λ_i are affine functions on $\mathcal{T}^l$ and verify $\lambda_i(\mathbf{a_j}) = \delta_{ij}$.

Proposition :
Let m, n and p three integer.

$$\int_{\mathcal{T}^l} \lambda_1^m \lambda_2^n \lambda_3^p \; dxdy = \frac{2\text{Area}(\mathcal{T}^l) m! n! p!}{(2+m+n+p)!}.$$

Example :

$$\int_{\mathcal{T}^l} \phi_2 \phi_3 dxdy = \int_{\mathcal{T}^l} \lambda_2 \lambda_3 dxdy = \frac{2\text{Area}(\mathcal{T}^l) 1!1!0!}{(2+1+1+0)!} = \frac{\text{Area}(\mathcal{T}^l)}{12}.$$

V. $\mathbb{P}_1$ finite element implementation : boundary conditions

We consider the problem

$$\text{Find } u \text{ such that} \qquad \begin{cases} -\Delta u + u = f & \text{in } \Omega, \\ u = g & \text{on } \Gamma_D, \\ \partial_n u = h & \text{on } \Gamma_N. \end{cases}$$

Variational formulation : We define $\widetilde{g}$ a lift of g, $w = u - \widetilde{g}$ and the space $V = \{u \in H^1(\Omega),\ u = 0 \text{ on } \Gamma_D\}$. We obtain

$$\begin{aligned} \text{Find } w \in V, \int_\Omega \nabla w \cdot \nabla v \ dx \ + \ \int_\Omega wv dx &= \int_\Omega fv dx + \int_{\Gamma_N} hv d\sigma \\ &- \int_\Omega \nabla \widetilde{g} \cdot \nabla v \ dx - \int_\Omega \widetilde{g} v dx \quad \forall v \in V. \end{aligned}$$

V. $\mathbb{P}_1$ finite element implementation : boundary conditions

We consider the problem

$$\text{Find } u \text{ such that} \qquad \begin{cases} -\Delta u + u = f & \text{in } \Omega, \\ u = g & \text{on } \Gamma_D, \\ \partial_n u = h & \text{on } \Gamma_N. \end{cases}$$

Discrete variational formulation : We define

$$V_h = \{v_h \in \mathcal{C}^0(\overline{\Omega}),\ v_h|_{\mathcal{T}_l} \in \mathcal{P}_1,\ \forall\ 1 \leq \cdots \leq N_t,\ v_h(\mathbf{a}_i) = 0\ \forall\ \mathbf{a}_i \in \Gamma_D\}$$

and obtain

$$\text{Find } w_h \in V_h, \int_\Omega \nabla w_h \cdot \nabla v_h\ dx \quad + \int_\Omega w_h v_h dx = \int_\Omega f v_h dx + \int_{\Gamma_N} h v_h d\sigma$$
$$- \int_\Omega \nabla \widetilde{g} \cdot \nabla v_h\ dx - \int_\Omega \widetilde{g} v_h dx \quad \forall v_h \in V_h.$$

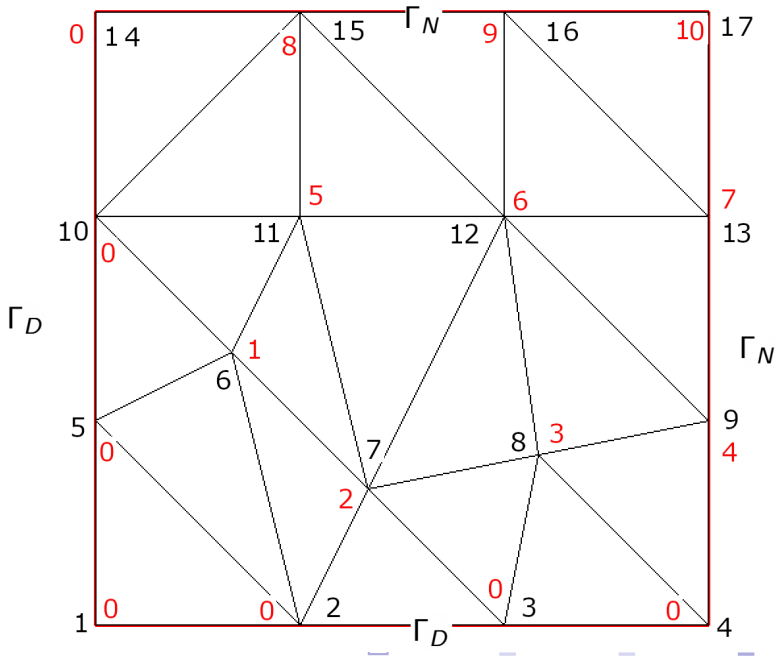
V. $\mathbb{P}_1$ finite element implementation : boundary conditions

Dirichlet treatment :

$$V_h = \{v_h \in \mathcal{C}^0(\overline{\Omega}),\ v_h|_{\mathcal{T}_l} \in \mathcal{P}_1,\ \forall\ 1 \leq \cdots \leq N_t,\ v_h(\mathbf{a}_i) = 0\ \forall\ \mathbf{a}_i \in \Gamma_D\}$$

We renumber the degree of freedom nodes with the creation of a table DOF :

```
n=0
DOF=0
For i = 1,...,N_v
    if a_i ∉ Γ_D then
        n=n+1
        DOF[i]=n
    end
end
```



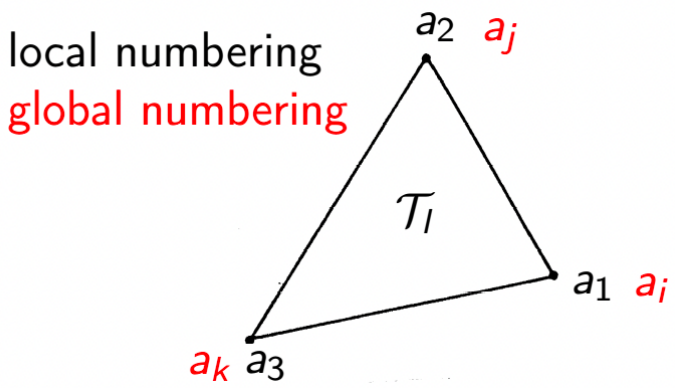
V. $\mathbb{P}_1$ finite element implementation : boundary conditions

Summary previous lecture :

$K = 0\,;\ M = 0\,;\ L = 0$
For $l = 1, \ldots, N_t$
    Build K^l_{elem}, M^l_{elem}, L^l_{elem}
    For $ldof1 = 1, \ldots, 3$
        $m = T[l, ldof1]$
        $L[m] = L[m] + L^l_{elem}[ldof1]$
        For $ldof2 = 1, \ldots, 3$
            $n = T[l, ldof2]$
            $K[m, n] = K[m, n] + K^l_{elem}[ldof1, ldof2]$
            $M[m, n] = M[m, n] + M^l_{elem}[ldof1, ldof2]$
        end
    end
end

We define elementary matrices and we assemble their elements into the global matrices using the triangle - vertex table T

$l = 1, \cdots, N_t$	i	j	k



V. $\mathbb{P}_1$ finite element implementation : boundary conditions

Dirichlet treatment :

```
K = 0 ; M = 0 ; L = 0
For l = 1, ..., N_t
    Build K^l_elem, M^l_elem, L^l_elem
    For ldof1 = 1, ..., 3
        mt = T[l, ldof1]
        m = DOF[mt]
        if m ≠ 0 then
            L[m] = L[m] + L^l_elem[ldof1]
            For ldof2 = 1, ..., 3
                nt = T[l, ldof2]
                n = DOF[nt]
                if n ≠ 0 then
                    K[m, n] = K[m, n] + K^l_elem[ldof1, ldof2]
                    M[m, n] = M[m, n] + M^l_elem[ldof1, ldof2]
                end if
            end
        end if
    end
end
```

V. $\mathbb{P}_1$ finite element implementation : boundary conditions

Dirichlet treatment :

To minimize the contribution of Dirichlet terms, we choose a lifting function localized around Γ_D :

$$\widetilde{g}(\mathbf{x}) = \sum_{j, \mathbf{a}_j \in \Gamma_D} g(\mathbf{a_j}) \phi_{\mathbf{j}}(\mathbf{x}).$$

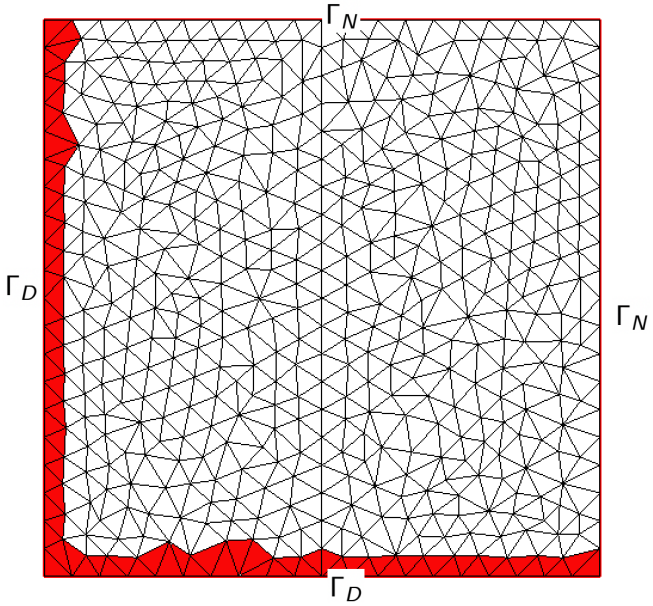


Figure – Support of the lifting function.

V. $\mathbb{P}_1$ finite element implementation : boundary conditions

Dirichlet treatment :

```
K = 0 ; M = 0 ; L = 0
For l = 1, ..., N_t
    Build K^l_elem, M^l_elem, L^l_elem
    For ldof1 = 1, ..., 3
        mt = T[l, ldof1]
        m = DOF[mt]
        if m ≠ 0 then
            L[m] = L[m] + L^l_elem[ldof1]
            For ldof2 = 1, ..., 3
                nt = T[l, ldof2]
                n = DOF[nt]
                if n ≠ 0 then
                    K[m, n] = K[m, n] + K^l_elem[ldof1, ldof2]
                    M[m, n] = M[m, n] + M^l_elem[ldof1, ldof2]
                else
                    L[m] = L[m] − (K^l_elem[ldof1, ldof2] + M^l_elem[ldof1, ldof2]) g[nt]
                end if
            end
        end if
    end
end
```

V. $\mathbb{P}_1$ finite element implementation : boundary conditions

Neumann treatment :

$$\int_{\Gamma_N} h v_h d\sigma$$

We use a boundary edge table Te

$$l = 1, \cdots, N_{ed,bd} \quad \begin{array}{|c|c|} \hline i & j \\ \hline \end{array}$$

and another elementary vector

$$L2^l_{elem} = \begin{pmatrix} \int_{\mathcal{E}^l} h\phi_1 \\ \int_{\mathcal{E}^l} h\phi_2 \end{pmatrix}.$$

```
For l = 1, ..., N_{ed,bd}
    Build L2^l_elem
    For ldof1 = 1, 2
        mt = Te[l, ldof1]
        m = DOF[mt]
        if m ≠ 0 then
            L[m] = L[m] + L2^l_elem[ldof1]
        end if
    end
end
```

Contents

1. Course introduction
2. Sobolev spaces
3. Elliptic problems
4. Variational approximation
5. $\mathbb{P}_1$ finite element implementation
6. Finite element error estimates

VI. Finite element error estimates

Reminder :

$$\text{(FV)} \qquad \text{Find } u \in V \text{ such that } a(u,v) = \ell(v),\ \forall v \in V.$$

$$\text{(F}V_h\text{)} \qquad \text{Find } u_h \in V_h \text{ such that } a(u_h,v_h) = \ell(v_h),\ \forall v_h \in V_h.$$

Lemma (Céa) :
There exists $C > 0$, independent of h, such that

$$\|u - u_h\|_V \leq C \inf_{v_h \in V_h} \|u - v_h\|_V.$$

Proposition :
Let $\mathcal{V}$ be a dense subspace of V and $r_h : \mathcal{V} \to V_h$ an operator such that

$$\forall v \in \mathcal{V}, \quad \lim_{h \to 0} \|v - r_h v\|_V = 0,$$

then the solution u_h of (FV_h) tends to the solution u of (FV) when $h \to 0$.

VI. Finite element error estimates : 1D $\mathbb{P}_1$ finite element

For simplicity of the presentation, we consider a uniform discretization : $x_i = (i-1)h$, for all $i = 1, \dots, N_v$ ($h = \frac{1}{N_v - 1}$).

Approximate space :

$$V_h = \{v_h \in \mathcal{C}^0([0,1]),\ v_h|_{[x_i, x_{i+1}]} \in \mathcal{P}_1,\ \forall\ 1 \leq i \leq N_v - 1\} \subset H^1(0,1).$$

We define the projector

$$r_h : \begin{cases} H^1(0,1) & \to V_h \\ v & \mapsto r_h v = \displaystyle\sum_{i=1}^{N_v} v(x_i)\phi_i \end{cases}.$$

Remark : in 1d, r_h can be defined for all $v \in V = H^1(0,1)$, thus it is not useful to introduce $\mathcal{V}$.

VI. Finite element error estimates : 1D $\mathbb{P}_1$ finite element

Proposition
For all $v \in H^1(0,1)$, $\lim_{h\to 0} \|r_h v - v\|_{H^1(0,1)} = 0$.

Corollary Let u and u_h the solution respectively of

$$\text{(FV)} \qquad \text{Find } u \in H^1(0,1) \text{ such that } a(u,v) = \ell(v),\ \forall v \in H^1(0,1).$$

$$\text{(F}V_h\text{)} \qquad \text{Find } u_h \in V_h \text{ such that } a(u_h,v_h) = \ell(v_h),\ \forall v_h \in V_h.$$

The $\mathbb{P}_1$ finite element converges : $\lim_{h\to 0} \|u - u_h\|_{H^1(0,1)} = 0$.

Corollary proof See Reminder Slide 51.

VI. Finite element error estimates : 1D $\mathbb{P}_1$ finite element

Proposition

For all $v \in H^1(0,1)$, $\lim_{h\to 0} \|r_h v - v\|_{H^1(0,1)} = 0$.

Moreover, there exists $C > 0$ such that, for all $v \in H^2(0,1)$,

$$\|r_h v - v\|_{H^1(0,1)} \leq Ch\|v''\|_{L^2(0,1)}.$$

Corollary Let u and u_h the solution respectively of

$$\text{(FV)} \qquad \text{Find } u \in H^1(0,1) \text{ such that } a(u,v) = \ell(v),\ \forall v \in H^1(0,1).$$

$$\text{(F}V_h\text{)} \qquad \text{Find } u_h \in V_h \text{ such that } a(u_h,v_h) = \ell(v_h),\ \forall v_h \in V_h.$$

The $\mathbb{P}_1$ finite element converges : $\lim_{h\to 0} \|u - u_h\|_{H^1(0,1)} = 0$.
Moreover, there exists $C > 0$ such that, if $u \in H^2(0,1)$,

$$\|u - u_h\|_{H^1(0,1)} \leq Ch\|u''\|_{L^2(0,1)}.$$

Corollary proof See Reminder Slide 51.

VI. Finite element error estimates : 1D $\mathbb{P}_1$ finite element

Proposition proof for the SECOND part (when $v \in H^2(0,1)$) :

To prove that there exists $C > 0$ such that, for all $v \in H^2(0,1)$,

$$\|r_h v - v\|_{H^1(0,1)} \leq Ch\|v''\|_{L^2(0,1)},$$

we will prove the two following L^2 estimates

$$\|r_h v - v\|_{L^2(0,1)} \leq h^2\|v''\|_{L^2(0,1)},$$

$$\|(r_h v)' - v'\|_{L^2(0,1)} \leq h\|v''\|_{L^2(0,1)}.$$

VI. Finite element error estimates : 1D $\mathbb{P}_1$ finite element

Let $v \in \mathcal{C}^\infty([0,1])$. In $[x_i, x_{i+1}]$, $r_h v(x) = v(x_i) + \frac{x-x_i}{x_{i+1}-x_i}\big(v(x_{i+1}) - v(x_i)\big)$.

It gives, using the mean value theorem,

$$\begin{aligned} r_h v(x) - v(x) &= v(x_i) - v(x) + \frac{x - x_i}{x_{i+1} - x_i}\big(v(x_{i+1}) - v(x_i)\big) \\ &= (x_i - x)v'(\xi_1) + (x - x_i)v'(\xi_2) \\ &= (x - x_i)\big(v'(\xi_2) - v'(\xi_1)\big) = (x - x_i)\int_{\xi_1}^{\xi_2} v''(t)dt. \end{aligned}$$

$$\begin{aligned} (r_h v(x) - v(x))^2 &= (x - x_i)^2\Big(\int_{\xi_1}^{\xi_2} v''(t)dt\Big)^2 \\ &\leq h^2\|v''\|^2_{L^1(x_i,x_{i+1})} \leq h^3\|v''\|^2_{L^2(x_i,x_{i+1})}. \end{aligned}$$

Integrating on $[x_i, x_{i+1}]$, we get

$$\|r_h v - v\|^2_{L^2(x_i,x_{i+1})} = \int_{x_i}^{x_{i+1}} (r_h v(x) - v(x))^2 dx \leq h^4\|v''\|^2_{L^2(x_i,x_{i+1})}.$$

VI. Finite element error estimates : 1D $\mathbb{P}_1$ finite element

$$\|r_h v - v\|^2_{L^2(x_i,x_{i+1})} = \int_{x_i}^{x_{i+1}} (r_h v(x) - v(x))^2 dx \leq h^4 \|v''\|^2_{L^2(x_i,x_{i+1})}.$$

Summing on each element, we obtain

$$\|r_h v - v\|_{L^2(0,1)} \leq h^2 \|v''\|_{L^2(0,1)}, \quad \forall v \in \mathcal{C}^\infty([0,1]).$$

Finally, by density (since $\mathcal{C}^\infty([0,1])$ is dense to $H^2(0,1)$),

$$\|r_h v - v\|_{L^2(0,1)} \leq h^2 \|v''\|_{L^2(0,1)}, \quad \forall v \in H^2(0,1).$$

VI. Finite element error estimates : 1D $\mathbb{P}_1$ finite element

End of lecture 10.

VI. Finite element error estimates : 1D $\mathbb{P}_1$ finite element

Similar approach for the derivative estimate :

Let $v \in \mathcal{C}^\infty([0,1])$. In $[x_i, x_{i+1}]$, $r_h v(x) = v(x_i) + \frac{x - x_i}{x_{i+1} - x_i}\big(v(x_{i+1}) - v(x_i)\big)$.

$$\begin{aligned}
(r_h v)'(x) - v'(x) &= \frac{v(x_{i+1}) - v(x_i)}{x_{i+1} - x_i} - v'(x) \\
&= \frac{1}{h}\int_{x_i}^{x_{i+1}} \big(v'(t) - v'(x)\big)\,dt = \frac{1}{h}\int_{x_i}^{x_{i+1}} \int_x^t v''(y)\,dy\,dt.
\end{aligned}$$

Using twice the Cauchy-Schwarz inequality,

$$\begin{aligned}
\big((r_h v)'(x) - v'(x)\big)^2 &= \frac{1}{h^2}\Big(\int_{x_i}^{x_{i+1}} \int_x^t v''(y)\,dy\,dt\Big)^2 \\
&\leq \frac{1}{h}\int_{x_i}^{x_{i+1}} \Big(\int_x^t v''(y)\,dy\Big)^2 dt \\
&\leq \frac{1}{h}\int_{x_i}^{x_{i+1}} (t - x)\|v''\|^2_{L^2(t,x)}\,dt \leq h\|v''\|^2_{L^2(x_i, x_{i+1})}.
\end{aligned}$$

VI. Finite element error estimates : 1D $\mathbb{P}_1$ finite element

$$\left((r_h v)'(x) - v'(x)\right)^2 \leq h\|v''\|^2_{L^2(x_i,x_{i+1})}.$$

Integrating on $[x_i, x_{i+1}]$ and summing on each element, we obtain

$$\|(r_h v)' - v'\|_{L^2(0,1)} \leq h\|v''\|_{L^2(0,1)}, \quad \forall v \in \mathcal{C}^\infty([0,1]).$$

Thus, by density,

$$\|(r_h v)' - v'\|_{L^2(0,1)} \leq h\|v''\|_{L^2(0,1)}, \quad \forall v \in H^2(0,1).$$

using the fact that $\|(r_h v)'\|_{L^2(0,1)} \leq \|v'\|_{L^2(0,1)}$ for all $v \in H^1(0,1)$.

Indeed, in $[x_i, x_{i+1}]$, we already wrote $(r_h v)'(x) = \frac{1}{h}\int_{x_i}^{x_{i+1}} v'(t)dt$. The usual procedure gives successively

$$|(r_h v)'(x)| \leq \frac{1}{\sqrt{h}}\|v'\|_{L^2(x_i,x_{i+1})}, \qquad \|(r_h v)'\|^2_{L^2(x_i,x_{i+1})} \leq \|v'\|^2_{L^2(x_i,x_{i+1})},$$

$$\|(r_h v)'\|^2_{L^2(0,1)} \leq \|v'\|^2_{L^2(0,1)}, \qquad \|(r_h v)'\|_{L^2(0,1)} \leq \|v'\|_{L^2(0,1)}.$$